

Projects:

Autonomous Fixed-Wing UAV – Teknofest Fighter UAV

Competition 2025: As part of JETDOSE, I worked on avionics integration, control systems, and real-time data handling for our autonomous fixed-wing UAV. Using Python and MAVSDK, I developed control functions, optimized communication protocols, and managed the Ground Control Station during flights. I also coordinated team efforts across subsystems, gaining experience in autonomous operations, mission logic, and UAV communication networks.



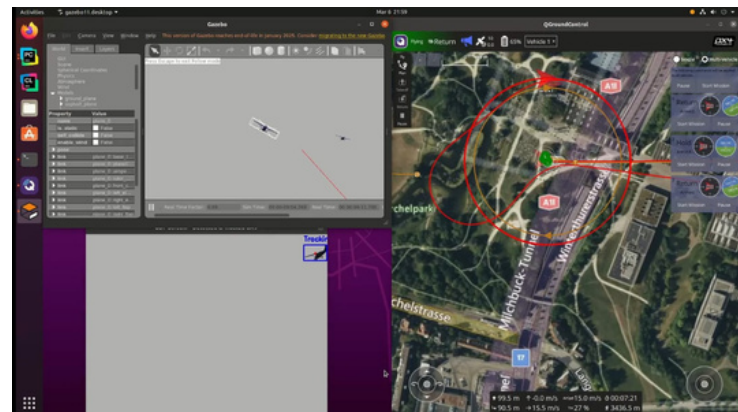
LIFELINE VTOL – Innovation & Invention Conference 3 (2025) - 2nd place

Our team developed a modular VTOL drone with vertical takeoff and efficient fixed-wing cruise, designed using 3D-printed structures and carbon-tube framing for scalability and field repair. The project achieved 2nd place overall and was ranked #1 in the UAV & Aerospace category, recognized as the top aerospace innovation at the competition. Applications include humanitarian aid, medical delivery, agriculture, and inspection.



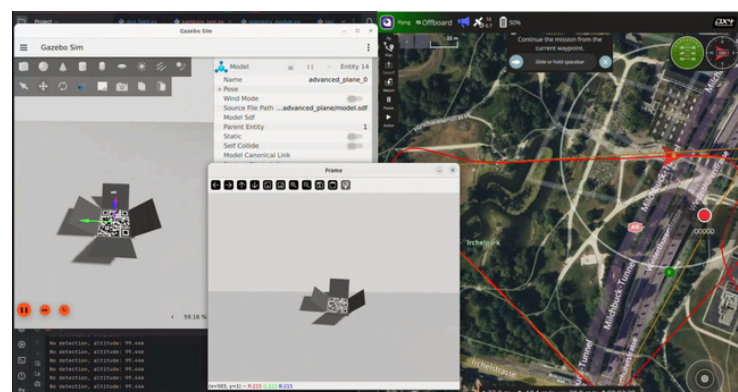
Dogfight mission developed in Python: I

developed a vision-based autonomous dogfight mission for a fixed-wing UAV. I used YOLOv8 together with sky segmentation to track targets from a simulated GStreamer video stream. By calculating tracking errors, I controlled the UAV's roll and pitch rates via MAVSDK Offboard mode. Real-time object detection and UAV control were executed using asyncio and multithreading. I simulated the mission control in the Gazebo environment.

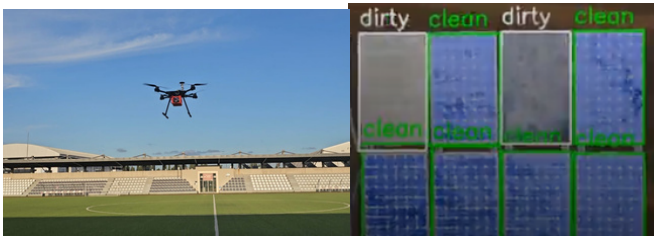


Kamikaze mission developed in Python: I

coded the kamikaze mission for the Teknofest Fighter UAV, primarily using MAVSDK and CV2. I then simulated the mission in the Gazebo environment.



Academic Research Project: I am currently contributing to an academic research project focused on developing a quadcopter capable of autonomously navigating solar farms. The UAV captures images of solar panels, analyzes them in real time, and identifies panels that are dirty or obstructed. The resulting data supports targeted maintenance and cleaning operations.



Autonomous Fixed-Wing UAV for Teknofest Fighter UAV Competition: I gained hands-on experience in both technical and project management domains. I contributed to the design and installation of UAV electronic systems as part of the avionics integration efforts. Using Python, I worked on UAV control systems and data processing tasks. To optimize performance and validate our control logic, I conducted flight simulations. In addition to my technical responsibilities, I took on the role of team organizer, ensuring effective collaboration and task distribution among team members. I also developed control functions, managed communication protocols, and operated the Ground Control Station (GCS) to monitor and control UAV operations in real time. As first-year Aerospace Engineering students, our 8-member team, ÜRK İHA, advanced to the semi-finals of Teknofest.



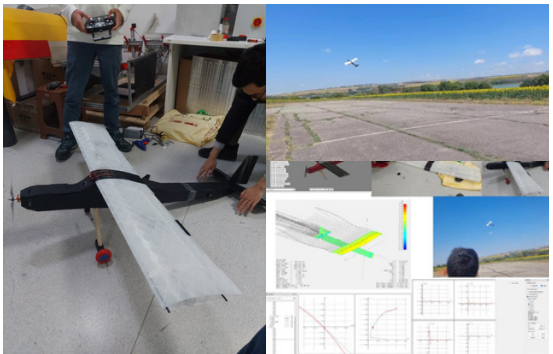
Autonomous Quadcopter for Teknofest Competition: I built a fully autonomous and stable quadcopter drone with 8 kg of thrust using the Pixhawk 4 Autopilot. This project was developed for the Teknofest Technology for Humanity Competition in 2019.



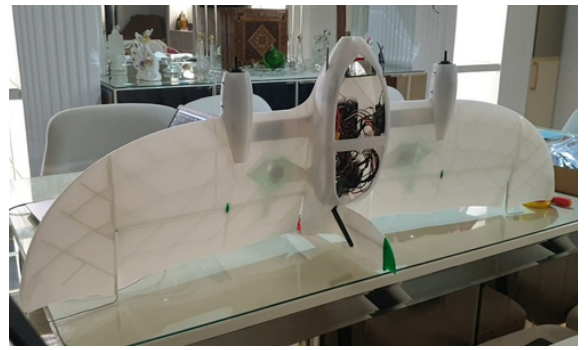
Autonomous Fixed-Wing Prototype for Teknofest Fighter UAV Competition: This project was developed as a prototype of the main system for the Teknofest Fighter UAV Competition. The airframe used was the X-UAV Mini Snow Goose, featuring a wingspan of 1.8 meters.



Assisted in Building a 1.5m Wingspan Fixed-Wing Aircraft: I contributed to the construction of a 1.5-meter wingspan RC trainer aircraft as part of my activities with the university's Uzay ve Roket Kulübü (Space and Rocket Club).



Autonomous Tailsitter VTOL Development: Involved in the development of an autonomous tailsitter VTOL UAV. The work includes mechanical assembly and electronics integration, with a particular emphasis on software development to enable aggressive maneuvers such as loops. Computer vision is also being integrated to enhance precision and control.



A330-343 Model Designed in Blender: I modeled the Airbus A330-343 aircraft in Blender, accurately capturing its characteristic design and structural details.

